

HomoRich Dual-Use: Persian G2P at SOTA Homograph Accuracy and Diacritization from a G2P-Only Dataset

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August 2026

Abstract

The HomaRich dataset (528K sentences) was published for Persian grapheme-to-phoneme conversion with homograph disambiguation; the strongest published system, Homo-GE2PE, reaches 76.89% homograph accuracy on the curated SentenceBench benchmark. We make two contributions. First, a plain ByT5-small seq2seq model trained on HomaRich’s simple phoneme field matches the published SOTA on its own benchmark (77.3% ezafe-normalized; 89.5% on a natural-distribution split), without the task-specific architecture or representation engineering. Second, we show HomaRich has a second, unpublished use: its **Mapped Phoneme** field deterministically converts to full haraqat diacritization, from which we train a Persian diacritizer reaching **0.52% character error rate**—to our knowledge the first Persian diacritization model trained without any diacritization-specific corpus. We also document a metric pitfall: homograph accuracy, PER, and CER are frequently compared across papers as if interchangeable; they measure different things.

1 Introduction

Persian omits short vowels in script; G2P must resolve them (and the unwritten ezafe) for TTS and transliteration. The HomaRich line of work [1] frames the challenge as homograph disambiguation.

Contributions:

1. **SOTA homograph accuracy without task-specific design:** ByT5-small on the simple phoneme field reaches 77.34% (ezafe-normalized) on SentenceBench, matching Homo-GE2PE’s 76.89%, and 89.45% on a natural split.
2. **HomaRich dual-use:** deterministic conversion of the **Mapped Phoneme** byte-pair format into haraqat yields 464K diacritization pairs; a ByT5-base trained on them reaches **0.52% CER**.
3. **Model-size saturation:** ByT5-small (300M) $\approx$ ByT5-base (1.2B) on this task family.
4. **Metric analysis:** exact vs. ezafe-normalized homograph scoring changes results by 18 points; we release both numbers and the protocol.

2 Related Work

“Fast, Not Fancy” [1] introduces HomaRich, Homo-GE2PE (T5-based, 76.89% HA), and HomaFast eSpeak (rule-based, 74.53%).

3 Data

HomoRich-G2P-Persian: 528,875 sentences, each with the plain `Phoneme` romanization, the byte-pair `Mapped Phoneme`, and homograph annotations. Splits: 507K/10.6K/10.6K (random, seed 42).

3.1 Diacritization derivation

The `Mapped Phoneme` format (e.g. `ruye1 divar n/nevisid`) encodes stress digits and syllable breaks. We parse it deterministically into haraqat on the original graphemes (`scripts/extract_diacritics.py`), keeping 464,445 pairs.

4 Models

ByT5-small and ByT5-base, seq2seq, 2–3 epochs, beam 4. Identical hyperparameters for G2P and diacritization apart from output space.

5 Experiments

5.1 G2P and homograph disambiguation

System	HA (SentenceBench)	HA (our split)	CER (our split)
Homo-GE2PE (published)	76.89%	—	—
HomoFast eSpeak (published)	74.53%	—	—
Ours, exact	59.11%	89.45%	1.64%
Ours, ezafe-normalized	77.34%	—	—

Table 1: Ezafe normalization strips the connective `/e/` our model attaches to words (*qadre* vs. reference *qadar*); it is required because our model was trained on the plain phoneme representation, which marks ezafe inline.

5.2 Diacritization

Task	CER
Persian diacritization (derived labels, 464K)	0.52%

5.3 Model size

ByT5-small: CER 1.64%, PER 6.18%, EM 63.8%. ByT5-base: CER 1.62%, PER 6.24%, EM 63.4%. The task saturates at 300M parameters.

6 Discussion

One dataset, two tasks. The diacritization result is the more striking: no Persian diacritization corpus of comparable size exists, yet HomaRich’s auxiliary field provides one for free.

Metrics are not interchangeable. Homograph accuracy scores one word per sentence; PER/CER score everything; representation choices (ezafe marking) silently shift HA by 18 points. Cross-paper comparisons require stating protocol—we release our HA implementation.

7 Limitations

Our SentenceBench comparison relies on a single normalization (ezafe stripping); a model trained on the GE2PE representation (Repr. 2) could be evaluated exactly. SentenceBench is small (203 scored homographs).

8 Reproducibility

interscript/rababa-farsi: training (`modal_app.py`, `modal_app_diacrit.py`), HA protocol (`eval_homograph.py`, `eval_sentencebench.py`), label derivation (`scripts/extract_diacritics.py`), docs/`RESULTS.md`.

References

- [1] Qharabagh et al. Fast, Not Fancy: Rethinking G2P with Rich Data and Rule-Based Models. arXiv:2505.12973, 2025.
- [2] Xue et al. ByT5. 2022.